

## Module 3: Data Link Layer and Wireless Networks



# Module 3: Data Link Layer and Wireless Networks

## Data-Link Layer

- Data Link Control (DLC)
- Multiple Access Protocols (MAC)
- Link-Layer Addressing
- Ethernet Protocol
- Connecting Devices

## Wireless Networks

- Wireless LANs
- Mobile IP

## Hands-On

- Datalink Provider Interface
- SOCK\_PACKET
- PF\_PACKET

# Introduction to Wired LANs

## TCP/IP and Lower Layers

- TCP/IP does **not define protocols** for the Data-Link and Physical Layers.
- It accepts any protocol at these layers that can provide services to the **Network Layer**.
- These layers are mainly associated with **LAN and WAN technologies**.

## Local Area Network (LAN)

- A LAN covers a **limited geographical area**.
- Examples:
  - Building
  - Campus
- Used for **resource sharing**.
- Most LANs today connect to a **WAN or the Internet**.

Chapter Focus: Wired Networks →  
Networks

Next Chapter: Wireless

# From LAN Technologies to Ethernet

## LAN Technologies

- Several LAN technologies were used during the **1980s and 1990s**.
- Each used a **media-access method** to share the communication medium.

| Technology | Method         |
|------------|----------------|
| Ethernet   | CSMA/CD        |
| Token Ring | Token passing  |
| Token Bus  | Token passing  |
| FDDI       | Token passing  |
| ATM LAN    | ATM technology |

## Why Ethernet Survived

- Almost all other LAN technologies **disappeared**.
- Ethernet could **evolve with increasing transmission-rate requirements**.
- Users could upgrade to a newer Ethernet generation instead of changing technology.

## Ethernet Evolution

| Generation          | Data Rate |
|---------------------|-----------|
| Standard Ethernet   | 10 Mbps   |
| Fast Ethernet       | 100 Mbps  |
| Gigabit Ethernet    | 1 Gbps    |
| 10 Gigabit Ethernet | 10 Gbps   |

**10 Mbps → 100 Mbps → 1 Gbps → 10 Gbps**

## Project 802

- Started in **1985** by the IEEE Computer Society.
- Developed standards for **intercommunication between equipment from different manufacturers.**
- Does **not replace** the OSI model or TCP/IP protocol suite.
- Specifies functions of the **Data-Link and Physical Layers** for major LAN protocols.

## IEEE 802 and TCP/IP

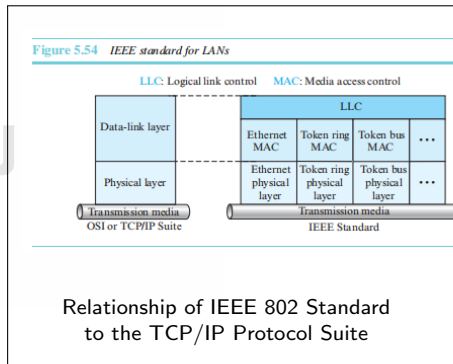


Figure 5.54

# Logical Link Control (LLC)

## IEEE 802 Data-Link Layer

IEEE divides the Data-Link Layer into two sublayers:

- **Logical Link Control (LLC)**
- **Media Access Control (MAC)**

## LLC Functions

- Flow control
- Error control
- Part of framing

## Role of LLC

- Provides a **single link-layer control protocol** for IEEE LANs.
- Makes the **MAC sublayer transparent**.
- Helps provide **interconnectivity between different LANs**.

### LLC

Flow Control + Error Control  
+ Part of Framing

# Media Access Control (MAC)

## Role of MAC

- Defines the **specific access method** for each LAN.
- Determines how devices **share the communication medium**.
- Handles **part of the framing function**.

## Access Methods

- Random access
- Controlled access
- Channelization

## Examples

|            |               |
|------------|---------------|
| LAN        | Access Method |
| Ethernet   | CSMA/CD       |
| Token Ring | Token passing |
| Token Bus  | Token passing |

### MAC

Specific medium-access method  
for a particular LAN

## 5.5.2 Standard Ethernet

### What is Standard Ethernet?

- Original Ethernet technology with a data rate of **10 Mbps**.
- Called **Standard Ethernet**.
- Although newer generations have appeared, several fundamental features have remained unchanged.
- Understanding Standard Ethernet helps us understand the other Ethernet generations.

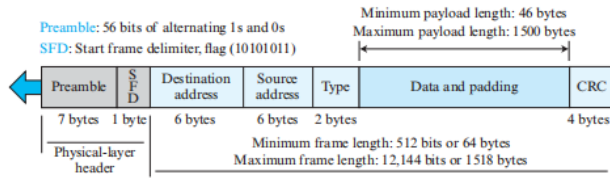
#### Main Topics

- 1 Frame Format
- 2 Connectionless and Unreliable Service
- 3 Frame Length
- 4 Addressing
- 5 Access Method
- 6 Efficiency
- 7 Implementations

**10 Mbps**

# Standard Ethernet Frame Format

Figure 5.55 Ethernet frame



Standard Ethernet Frame Format

- **Preamble** — 7 bytes
- **SFD** — 1 byte
- **Destination Address** — 6 bytes
- **Source Address** — 6 bytes
- **Type** — identifies upper-layer protocol
- **Data** — 46–1500 bytes
- **CRC** — error detection

# Ethernet Frame: Preamble and SFD

## Preamble

- Size: **7 bytes = 56 bits**.
- Contains alternating **0s and 1s**.
- Alerts the receiver that a frame is coming.
- Helps the receiver **synchronize its clock**.
- Added at the **Physical Layer**.

**Preamble = Synchronization**

## Start Frame Delimiter (SFD)

- Size: **1 byte**.
- Pattern: **10101011**.
- Signals the **beginning of the frame**.
- Last two bits are **11**.
- Also added at the **Physical Layer**.

**SFD = Start of Frame**

# Ethernet Frame: Address, Type, Data and CRC

## Destination Address (DA)

- **6 bytes (48 bits).**
- Identifies the destination station(s).

## Source Address (SA)

- **6 bytes (48 bits).**
- Identifies the sender.

## Type

- Identifies the upper-layer protocol.
- Examples: **IP, ARP, OSPF.**
- Used for **multiplexing/demultiplexing.**

## Data

- Carries the upper-layer packet.
- Minimum: **46 bytes.**
- Maximum: **1500 bytes.**
- Data below 46 bytes is **padded.**

## CRC

- Last field of the frame.
- Uses **CRC-32** for error detection.
- Corrupted frames are **discarded.**

# Ethernet Service: Connectionless and Unreliable

## Connectionless Service

- Each frame is sent **independently**.
- No connection establishment phase.
- No connection termination phase.
- Sender transmits whenever it has a frame.

## Unreliable Service

- Ethernet does not guarantee frame delivery.
- A corrupted frame is **discarded silently**.
- Higher-layer protocols are responsible for recovery.
- With TCP, retransmission can recover lost data.
- With UDP, recovery may depend on the application.

**Ethernet: Connectionless + Unreliable**

# Ethernet Frame Length

## Minimum Frame Length

- **64 bytes = 512 bits.**
- Header + trailer = **18 bytes.**
- Minimum data:

$$64 - 18 = \mathbf{46 \text{ bytes}}$$

- Padding is added when data is less than 46 bytes.

## Maximum Frame Length

- **1518 bytes** excluding Preamble and SFD.
- Header + trailer = **18 bytes.**
- Maximum payload:

$$1518 - 18 = \mathbf{1500 \text{ bytes}}$$

$$\mathbf{64 \text{ bytes} \leq \text{Frame} \leq 1518 \text{ bytes}}$$

# Why Does Ethernet Restrict Frame Length?

## Minimum Length

- Required for the correct operation of **CSMA/CD**.
- Sender must remain transmitting long enough to detect a possible collision.

Minimum size → Collision detection

## Maximum Length

- Early Ethernet used expensive memory.
- Maximum frame size helped reduce **buffer requirements**.
- Prevents one station from **monopolizing the shared medium**.

Maximum size → Fair sharing

## Ethernet Address

- Every Ethernet station has a **Network Interface Card (NIC)**.
- The NIC provides the station with a **link-layer address**.
- Ethernet address size: **6 bytes = 48 bits**.

## Address Representation

47:20:1B:2E:08:EE

- Written in hexadecimal notation.
- Six hexadecimal byte pairs.
- Each byte contains **8 bits**.

**Ethernet MAC Address = 48 bits**

# Transmission of Address Bits

## How Are Address Bits Transmitted?

- Bytes are transmitted from **left to right**.
- Within each byte, bits are transmitted from **least significant bit (LSB) first**.
- The most significant bit (MSB) is transmitted last.
- Therefore, the bit identifying an address as **unicast or multicast arrives first**.

### Example 5.18

47:20:1B:2E:08:EE

Hexadecimal: 47 20 1B 2E 08 EE

Transmitted bit order: 11100010 00000100 11011000 01110100 00010000 01110111

# Unicast, Multicast and Broadcast Addresses

## Unicast

- Source address is always unicast.
- Destination is one station.
- LSB of first byte = **0**.

## Multicast

- Destination represents a group.
- LSB of first byte = **1**.
- Group members accept the frame.

## Broadcast

- Destination is **all stations**.
- Address contains 48 ones.
- Written as: FF:FF:FF:FF:FF:FF

**Figure 5.56** *Unicast and multicast addresses*



## Example 5.19: Address Classification

### Rule

- Look at the **second hexadecimal digit** from the left.
- **Even** → Unicast
- **Odd** → Multicast
- All digits are **F** → Broadcast

| Destination Address | Type      |
|---------------------|-----------|
| 4A:30:10:21:10:1A   | Unicast   |
| 47:20:1B:2E:08:EE   | Multicast |
| FF:FF:FF:FF:FF:FF   | Broadcast |

**Example:** 4A → A = 1010 (even) → Unicast

# Standard Ethernet: Who Keeps the Frame?

**All stations receive the frame — but who keeps it?**

## Unicast

### ONE

**One intended recipient**

- All stations receive the frame.
- The intended recipient **keeps and processes** it.
- Others **discard** it.

## Multicast

### GROUP

**Selected group of recipients**

- All stations receive the frame.
- **Group members** keep and process it.
- Others **discard** it.

## Broadcast

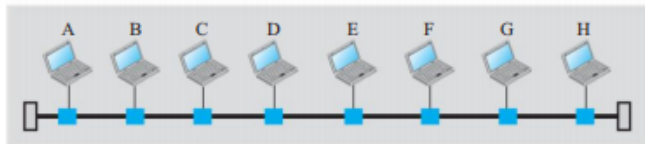
### EVERYONE

**All receiving stations**

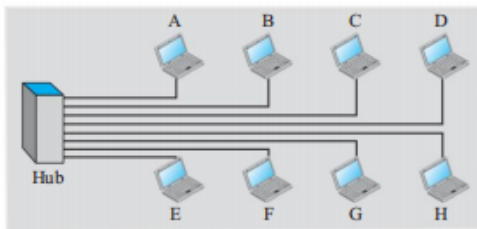
- All stations except the sender receive the frame.
- All receiving stations **keep and process** it.

**Key Idea:** All stations may receive the frame; the **destination address determines who keeps it.**

**Figure 5.57** *Implementation of standard Ethernet*

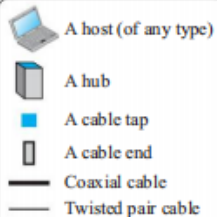


a. A LAN with a bus topology using a coaxial cable



b. A LAN with a star topology using a hub

**Legend**



# Standard Ethernet Access Method: CSMA/CD

## Access Method

- Standard Ethernet uses **CSMA/CD**.
- Uses the **1-persistent** method.
- A station first performs **carrier sensing**.
- If the medium is idle, transmission begins.
- If the medium is busy, the station waits until the medium becomes idle.

**Carrier Sense → Transmit → Monitor for Collision**

## Case 1: No Collision

- Station transmits and continuously monitors the medium.
- If no collision is detected during the first **512 bits**, transmission is considered successful.
- Station can discard its buffered copy.

## Case 2: Collision

- Collision detected before 512 bits.
- Station aborts transmission.
- Sends a **48-bit jam signal**.
- Increases the attempt count  $K$ .
- Performs backoff before retransmission.

**512 bits = Collision Detection Window**

# Why Does Ethernet Use 512 Bits?

## At 10 Mbps

$$\text{Transmission time} = \frac{512 \text{ bits}}{10 \text{ Mbps}} = 51.2 \mu\text{s}$$

## Propagation Consideration

- Signal propagation speed:

$$2 \times 10^8 \text{ m/s}$$

- During  $51.2 \mu\text{s}$ , the first bit can travel a significant distance and return after a collision.
- Therefore, if no collision is detected during the first **512 bits**, the sender assumes that no collision will occur.

**512-bit minimum frame size supports CSMA/CD collision detection.**

# CSMA/CD: Binary Exponential Backoff

## After a Collision

- Station sends a **48-bit jam signal**.
- Increment the collision count  $K$ .
- If  $K = 15$ , the station aborts the attempt.
- If  $K < 15$ , the station waits for a random backoff time.

| Collision | Random Number Range |
|-----------|---------------------|
| $K = 1$   | 0-1                 |
| $K = 2$   | 0-3                 |
| $K = 3$   | 0-7                 |
| $K = 4$   | 0-15                |

$$0 \leq R \leq 2^K - 1$$

# Efficiency of Standard Ethernet

## Efficiency

The efficiency of Ethernet is the ratio of the time used by a station to send data to the time the medium is occupied by the station.

$$\text{Efficiency} = \frac{1}{1 + 6.4a}$$

where

$$a = \frac{\text{Propagation Delay}}{\text{Transmission Delay}}$$

- Smaller  $a \rightarrow$  higher efficiency.
- Shorter medium or longer frame  $\rightarrow$  higher efficiency.
- Ideal case:  $a = 0$ , efficiency = 1.

## Example 5.20: Ethernet Efficiency

### Given

- Transmission rate = **10 Mbps**
- Medium length = **2500 m**
- Frame size = **512 bits**
- Propagation speed =  $2 \times 10^8$  m/s

$$\text{Propagation Delay} = \frac{2500}{2 \times 10^8} = 12.5 \mu s$$

$$a = \frac{12.5}{51.2} \approx 0.24$$

$$\text{Transmission Delay} = \frac{512}{10^7} = 51.2 \mu s$$

$$\text{Efficiency} = \frac{1}{1 + 6.4(0.24)} \approx 39\%$$

$$\text{Efficiency} \approx 39\%$$

# Standard Ethernet Implementations

## Popular Implementations

- Standard Ethernet defined several implementations.
- Four implementations became popular during the **1980s**.

## Understanding 10BaseX

- **10** → data rate = 10 Mbps.
- **Base** → baseband digital signaling.
- **X** → cable length or cable type.

**All Standard Ethernet implementations use Manchester encoding.**

# 10BaseX Nomenclature

| Part        | Meaning                    |
|-------------|----------------------------|
| <b>10</b>   | Data rate = 10 Mbps        |
| <b>Base</b> | Baseband digital signal    |
| <b>X</b>    | Cable length or cable type |

## Cable-Length Examples

- **5** → approximately 500 m
- **2** → approximately 185 m

## Cable-Type Examples

- **T** → UTP cable
- **F** → Fiber-optic cable

**Standard Ethernet uses baseband signaling and Manchester encoding.**

# Table 5.6: Standard Ethernet Implementations

| Implementation | Medium     | Length | Encoding   |
|----------------|------------|--------|------------|
| 10Base5        | Thick coax | 500 m  | Manchester |
| 10Base2        | Thin coax  | 185 m  | Manchester |
| 10Base-T       | UTP        | 100 m  | Manchester |
| 10Base-F       | Fiber      | 2000 m | Manchester |

## Remember

**All Standard Ethernet implementations use Manchester encoding.**

10Base5 → 500 m

10Base2 → 185 m

10Base-T → 100 m

10Base-F → 2000 m

## 5.5.3 Fast Ethernet (100 Mbps)

### Why Fast Ethernet?

- Ethernet increased its transmission rate from **10 Mbps to 100 Mbps**.
- The new generation was called **Fast Ethernet**.

### Compatibility with Standard Ethernet

- The **MAC sublayer and Frame format was left unchanged**.
- Minimum and maximum frame sizes remained unchanged.
- However, the **access method and implementation** had to be reconsidered.

### Key Idea

**10 Mbps → 100 Mbps**

10 times higher transmission rate

### What had to change?

Features depending on the transmission rate had to be reconsidered, particularly:

- CSMA/CD operation
- Network length
- Physical implementation

# Fast Ethernet: Access Method

## Why Reconsider CSMA/CD?

CSMA/CD depends on:

1. Transmission rate
2. Minimum frame size
3. Maximum network length

If the minimum frame size remains **512 bits**:

- Fast Ethernet transmits the frame **10 times faster**.
- Collision must therefore be detected **10 times sooner**.
- Propagation speed remains unchanged.
- Hence, the maximum network length must become **10 times shorter**.

| Ethernet          | Data Rate | Maximum Length |
|-------------------|-----------|----------------|
| Standard Ethernet | 10 Mbps   | 2500 m         |
| Fast Ethernet     | 100 Mbps  | 250 m          |

# Fast Ethernet: Two Solutions

## Solution 1: Passive Hub + Star

- The **bus topology** is dropped.
- A **passive hub** and **star topology** are used.
- Maximum network size is reduced to **250 m**.
- This approach is retained for **compatibility with Standard Ethernet**.

## Solution 2: Switch + Full Duplex

- A **link-layer switch with buffer** is used.
- Each host has a **full-duplex connection**.
- The medium becomes private for each host.
- Hosts do not compete for the medium.
- Therefore, **CSMA/CD is not required**.

## Transformation

Shared medium      →      Many point-to-point media

# Fast Ethernet: How the Switch Works

## Frame Processing

The link-layer switch:

- ① Receives a frame from the source host.
- ② Stores the frame in a **buffer (queue)**.
- ③ Checks the **destination address**.
- ④ Sends the frame through the corresponding interface.

## Full-Duplex Operation

- Each host has a private connection to the switch.
- Transmission and reception can occur simultaneously.
- A destination host can receive a frame while sending a frame to another station.
- No contention occurs between hosts.

**No shared medium  $\Rightarrow$  No competition  $\Rightarrow$  No CSMA/CD**

# Fast Ethernet: Autonegotiation

## What is Autonegotiation?

Autonegotiation is a feature introduced in Fast Ethernet that allows two devices to negotiate their:

- **Operating mode**
- **Data rate**

## Purpose

- Allows devices with different capabilities to communicate.
- A device can advertise its available capabilities.
- The devices select a compatible operating mode and rate.

## Example

**10 Mbps device**



**100 Mbps device**

Communication is possible if the 100 Mbps device can operate at the lower rate.

# Fast Ethernet: Physical Implementations

## Classification

Fast Ethernet implementations at the physical layer are categorized as:

### Two-Wire

- **100Base-TX**
  - Shielded Twisted Pair (STP)
- **100Base-FX**
  - Fiber-optic cable

### Four-Wire

- **100Base-T4**
- Designed only for **Unshielded Twisted Pair (UTP)**

## Table 5.7 — Fast Ethernet Implementations

| Implementation    | Medium | Length | Wires | Encoding     |
|-------------------|--------|--------|-------|--------------|
| <b>100Base-TX</b> | STP    | 100 m  | 2     | 4B5B + MLT-3 |
| <b>100Base-FX</b> | Fiber  | 185 m  | 2     | 4B5B + NRZ-I |
| <b>100Base-T4</b> | UTP    | 100 m  | 4     | Two 8B/6T    |

### Remember

**Fast Ethernet = 100 Mbps**

MAC sublayer and frame format remain compatible with Standard Ethernet, while physical implementation supports the higher rate.

## 5.5.4 Gigabit Ethernet (1000 Mbps)

### Introduction

- Data rate: **1000 Mbps = 1 Gbps.**
- IEEE standard: **802.3z.**

### Main Goals

Gigabit Ethernet aimed to:

- Increase the data rate to **1 Gbps.**
- Keep the **48-bit address length** unchanged.
- Keep the **frame format** unchanged.
- Keep the **minimum and maximum frame lengths** unchanged.

### Evolution of Ethernet

**10 Mbps**  
Standard Ethernet



**100 Mbps**  
Fast Ethernet



**1000 Mbps**  
Gigabit Ethernet

### Important

A major goal in Ethernet evolution was to keep the **MAC sublayer unchanged**. However, achieving 1 Gbps made this no longer completely possible.

# Gigabit Ethernet: MAC Sublayer

## Two Approaches for Medium Access

Gigabit Ethernet has two distinctive approaches:

- **Half-duplex**
- **Full-duplex**

**Almost all implementations use full-duplex mode.**

## Full-Duplex Architecture

- A **central switch** is connected to all computers or other switches.
- Each input port of the switch has a **buffer**.
- Frames are stored until they are ready for transmission.

## Frame Forwarding

- The switch examines the **destination address**.
- It sends the frame through the port connected to the destination.
- Different hosts do not compete for the same medium.

# Gigabit Ethernet: Cable Length

## Why No CSMA/CD?

- The switch provides a separate connection to each host.
- Frames are forwarded using the destination address.
- Therefore, there is **no collision**.
- CSMA/CD is therefore **not used**.

## Effect on Cable Length

Since collisions do not occur:

- Cable length is no longer determined by the **collision detection process**.
- Instead, it is determined by **signal attenuation in the cable**.

**No collision detection constraint**



**Cable length depends on signal attenuation**

# Gigabit Ethernet: Implementation

## Physical-Layer Implementations

Table 5.8 summarizes the Gigabit Ethernet implementations.

- **S-W** = Short Wave
- **L-W** = Long Wave
- Encoding is discussed in Chapter 7.

| Implementation     | Medium | Wave | Length | Wires | Encoding     |
|--------------------|--------|------|--------|-------|--------------|
| <b>1000Base-SX</b> | Fiber  | S-W  | 550 m  | 2     | 8B/10B + NRZ |
| <b>1000Base-LX</b> | Fiber  | L-W  | 5000 m | 2     | 8B/10B + NRZ |
| <b>1000Base-CX</b> | STP    | –    | 25 m   | 2     | 8B/10B + NRZ |
| <b>1000Base-T4</b> | UTP    | –    | 100 m  | 4     | 4D-PAM5      |

# Gigabit Ethernet — Quick Review

## Remember

- Data rate: **1 Gbps**
- IEEE standard: **802.3z**
- Almost all implementations use **full-duplex**.
- **CSMA/CD is not used.**

## Main Implementations

- 1000Base-SX → Fiber, 550 m
- 1000Base-LX → Fiber, 5000 m
- 1000Base-CX → STP, 25 m
- 1000Base-T4 → UTP, 100 m

## Core Concept

**Gigabit Ethernet uses switched full-duplex communication, eliminating collisions and the need for CSMA/CD.**

### 5.5.6 Virtual LANs

# What is a VLAN?

- A station is traditionally considered part of a LAN if it **physically belongs** to that LAN.
- LAN membership is therefore based on **geography and physical wiring**.
- A **Virtual Local Area Network (VLAN)** is a LAN configured by **software rather than physical wiring**.

## Why VLAN?

- A work group may contain stations connected to different physical locations.
- Moving a person between work groups should not require physical rewiring.
- VLAN membership can be changed through software.

## Membership is logical, not physical

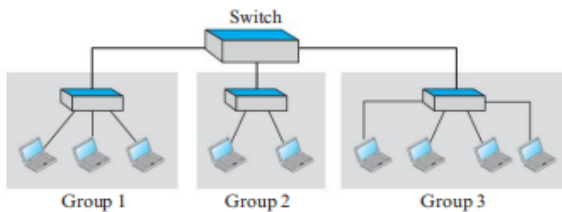
## Core Idea

**Physical LAN**     $\longrightarrow$     **Logical LAN**  
                               **(VLAN)**

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**Figure 5.58** *A switch connecting three LANs*

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# VLAN Example: Logical Work Groups

## Switched LAN

- Ten stations in an engineering firm are grouped into three LANs.
- The groups represent different work groups.
- The LAN is configured according to these groups.
- If two engineers move to another group, the physical network configuration would need to be changed.

## Problem

Physical movement between work groups requires **rewiring** and repeated configuration changes.



**Physical work-group change  $\Rightarrow$  Physical network change**

# VLAN: Logical Segmentation

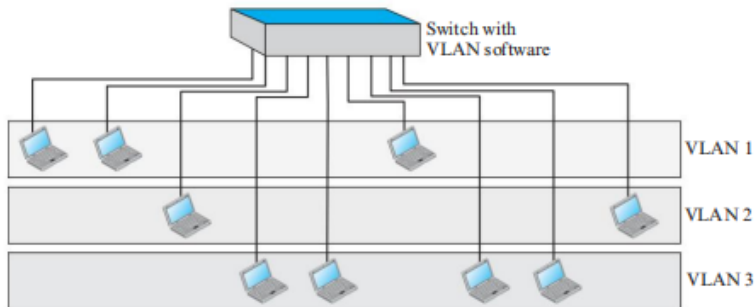
## How VLAN Solves the Problem

- A LAN can be divided into several **logical LANs called VLANs**.
- Each VLAN represents a **work group**.
- Group membership is defined by **software, not hardware**.
- A station can be logically moved from one VLAN to another.

## Broadcast Behavior

- Members of a VLAN receive broadcasts sent to that VLAN.
- A station moved from VLAN 1 to VLAN 2 receives VLAN 2 broadcasts, but not VLAN 1 broadcasts.

**Figure 5.59** *A switch using VLAN software*



# VLANs Across Multiple Switches

## VLAN Across Physical LANs

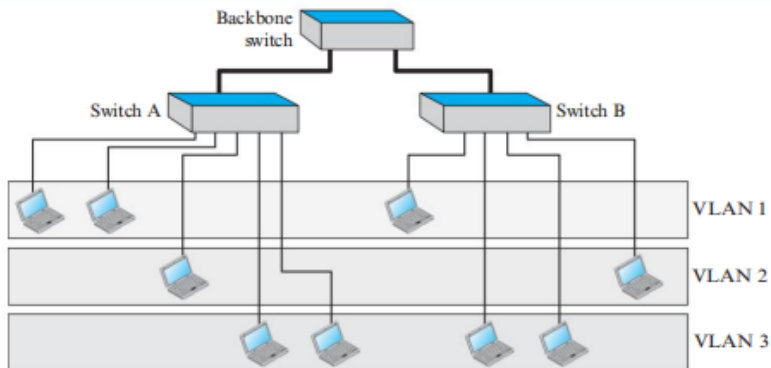
- VLAN technology can group stations connected to **different switches**.
- Stations from switches A and B can belong to the same VLAN.
- Useful when an organization has **two separate buildings**.
- People in different buildings can belong to the same work group.

## Broadcast Domain

A VLAN defines a **broadcast domain**.

**Stations in a VLAN communicate as though they belong to one physical segment.**

**Figure 5.60** *Two switches in a backbone using VLAN software*



# VLAN Membership

## Characteristics Used for Membership

Vendors may use one or more of the following characteristics:

### 1. Interface / Port Numbers

- VLAN membership can be based on **switch interface numbers**.
- Example: Ports 1, 2, 3, and 7 → VLAN 1.

### 2. MAC Addresses

- Uses the **48-bit MAC address**.
- Specific stations can be assigned to a VLAN.

### 3. IP Addresses

- Uses the **32-bit IP address**.
- Stations with selected IP addresses can belong to the same VLAN.

### 4. Multicast IP Addresses

- Uses the **multicast IP address**.
- IP-layer multicasting is translated to data-link-layer multicasting.

# VLAN Configuration

## Manual

- Administrator manually assigns stations to VLANs.
- Migration is also done manually.
- Configuration is logical, not physical.

## Automatic

- Stations are automatically connected or disconnected from VLANs.
- Membership is based on administrator-defined criteria.
- Example: **project number**.

## Semiautomatic

- Between manual and automatic configuration.
- Initializing is usually done manually.
- Migration is done automatically.

# Communication Between Switches

## The Problem

In a multi-switched backbone, each switch must know the VLAN membership of stations connected to other switches.

### 1. Table Maintenance

- Switch records station membership when a broadcast frame is received.
- Switches periodically exchange their tables.

### 2. Frame Tagging

- An extra header is added to the MAC frame.
- The tag identifies the **destination VLAN**.
- Receiving switches use the tag to identify the VLAN.

### 3. TDM

- The trunk between switches is divided into **time-shared channels**.
- Each VLAN is assigned a channel.
- Receiving switch identifies the VLAN from the channel.

# IEEE 802.1Q Standard

## IEEE Standard

- In **1996**, the IEEE 802.1 subcommittee passed the **802.1Q** standard.
- It defines the format for **frame tagging**.
- It also defines the format used in **multi-switched backbones**.

## Purpose

- Enables the use of **multivendor equipment** in VLANs.
- Provides a basis for further standardization of VLAN-related issues.
- Most vendors have accepted the standard.

## IEEE 802.1Q

Standard for VLAN frame tagging

# Advantages of VLANs

## Cost and Time Reduction

- Physical reconfiguration is costly and time-consuming.
- Stations can be moved between groups using **software**.
- No physical rewiring is required.

## Virtual Work Groups

- Stations can be grouped according to **work/project**.
- Members can communicate through broadcasts.
- Members need not belong to the same department or physical LAN.

## Security

- VLANs provide an **extra measure of security**.
- Broadcast messages sent within a group are not received by users in other VLAN groups.

## Final Idea

**VLAN = Logical grouping of stations into broadcast domains, independent of their physical location.**